

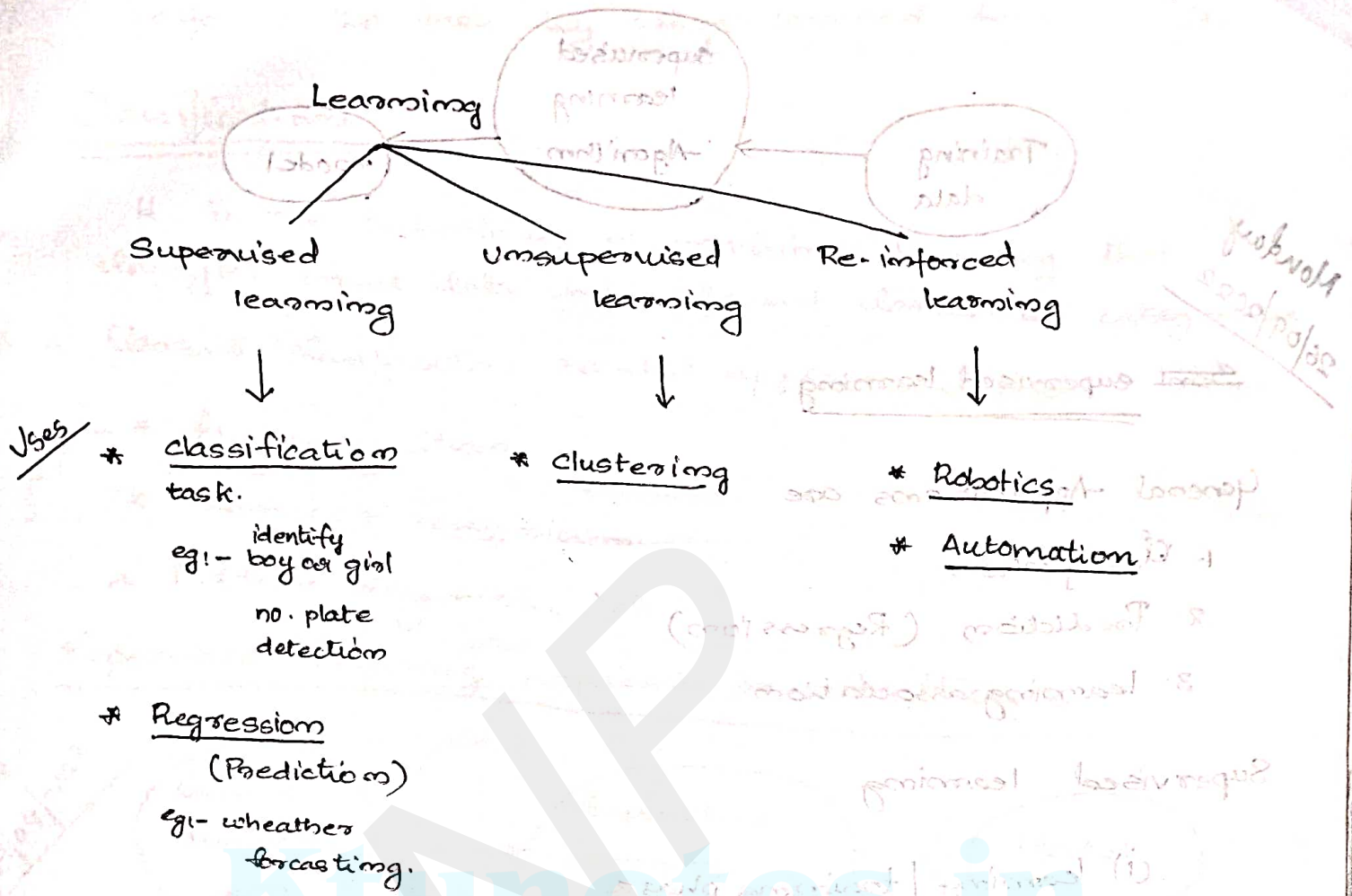


KTU NOTES

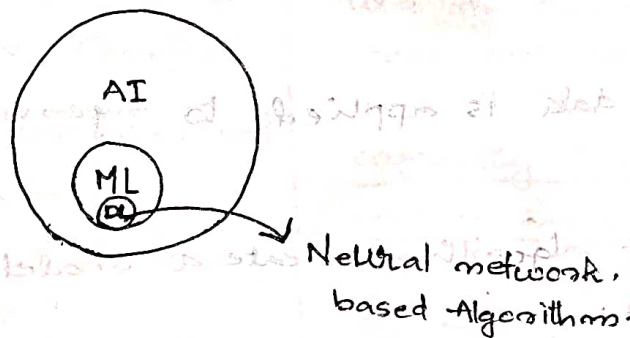
The learning companion.

**KTU STUDY MATERIALS | SYLLABUS | LIVE
NOTIFICATIONS | SOLVED QUESTION PAPERS**

Website: www.ktunotes.in



Relation between AI and ML



SUPERVISED LEARNING

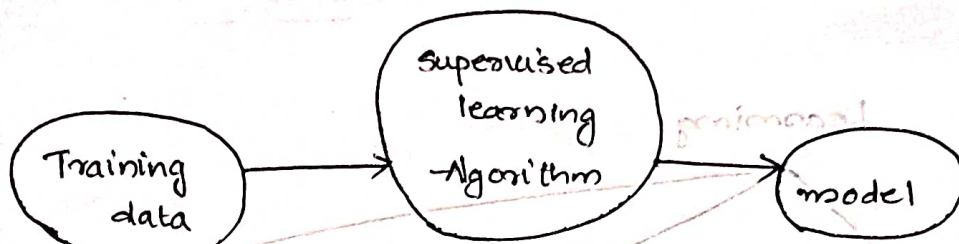
- It involves ~~supervised~~ learning / training phase and testing phase

Applications

- * classification
- * Regression (Prediction)
- * learning association. [data makes the relation strong]

Learning phase (classification)

MODULE-1



Monday
26/09/2022

Semi supervised learning.

General Applications are

1. Classification
2. Prediction (Regression)
3. Learning association

Supervised learning

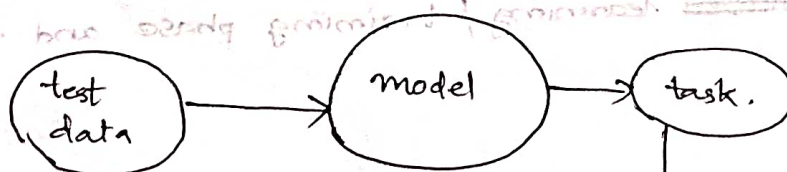
- (i) learning / training phase
- (ii) Testing Phase.

(i) Training phase

Labelled training data is applied to supervised learning Algorithms.

Then the learning algorithms create a model. that model performs the tasks.

(ii) Testing phase



- classification
- Regression

Test data is applied to the model. Then the model performs the task by using learned knowledge.

Classifications

It is the technique in machine learning that classifies input data into different classes, or categories.

Various classification related applications are:

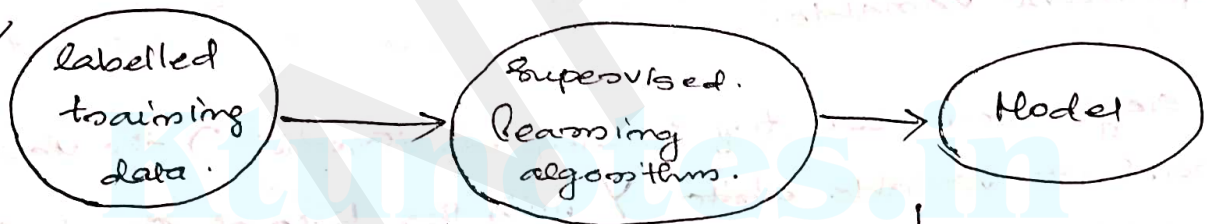
* face recognition.

* character recognition.

* Medical diagnosis ... etc.

Supervised learning approach for classification.

Tue
24/09/2022



* Labeled training data contains "instances of object" to be classified

* Each ~~is~~ instance will be in the form
 <features of object, o/p class>

training pair.

eg! - classifying customers as high risk or low risk.

eg! -

	Income	expenditure	class
c1	25,000	15,000	Low
c2	25,000	24,000	High.

* Learning algorithm uses the training data and create a model that correctly classifies each instance into the corresponding o/p class.

→ Training algorithm.

* Decision tree (ID3) algorithm.

* (SVM) Support Vector Machine.

* Bayesian classifier

* NN (ANN)

* K-nearest Neighbours

* Model performs classification

* DT → Tree

* (ANN)

Artificial Neural Network → collection of Artificial Neurons

* (SVM) → (line/ plane equations)

Seminar.

September 29th.

→ Topic → Guide.

Oct - 11 5 students.

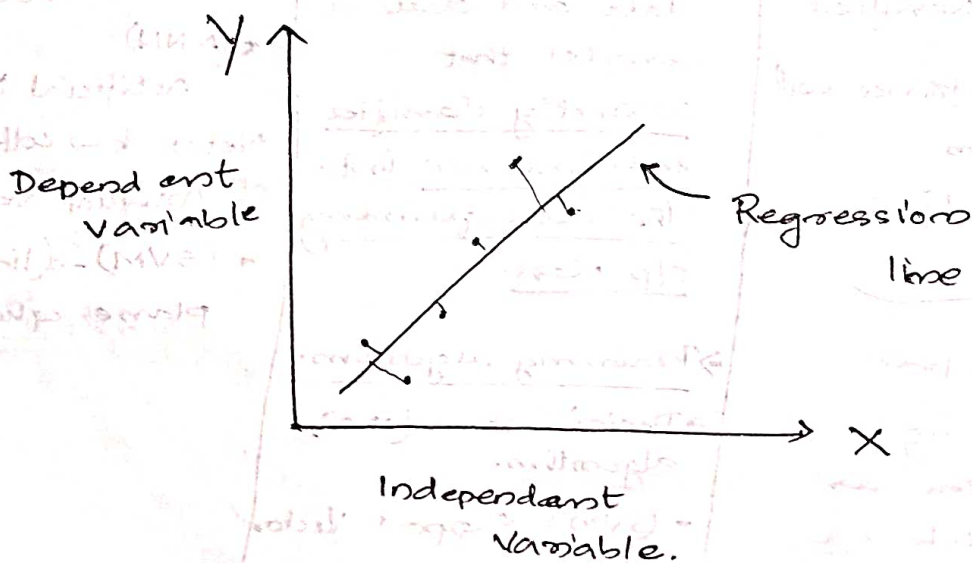
Testing phase

Test data is given [data ~~is~~ ^{not} used for training] to the model. Then the model performs classification.

Supervised learning Approach for Regression [Predictions]

Regression is a statistical method that helps to analyze and understand the relationship between two or more variables. These variables are called dependant variable and Independent variable.

- * In regression, we try to regress (Predict) the value of dependant variable ~~able~~ Y with the help of the independent variable X or variables. That is we try to understand how the value of Y changes with respect to change in X .

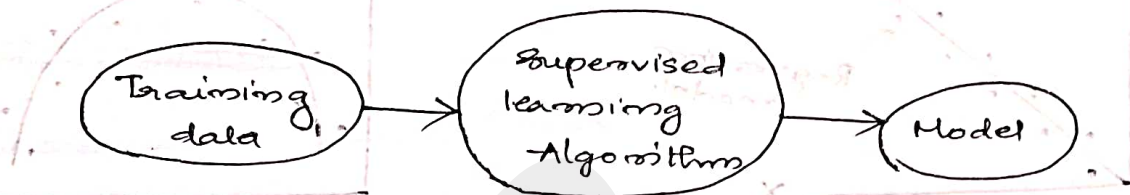


Regression line just best describe the relationship between X and Y

Applications

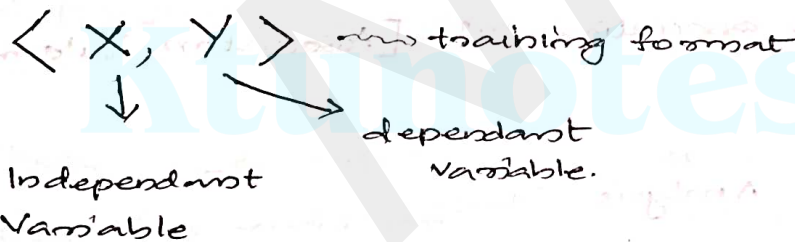
- * Weather forecasting.
- * In the stock market for stock price predictions.
- * Predicting the price of the used car.

Machine learning Approach



Training phases:

During the training phase past data about a problem to be solved is used.



This training data is fed to a Supervised learning algorithm then the learning algorithm creates a regression model.

The model correctly predict the relationship b/w X and Y and o/p the predictions

For example:

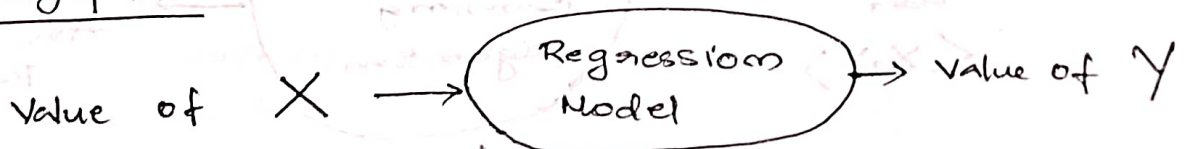
$$Y = wX + w_0$$

$$Y = mx + c$$

Eq. for the line

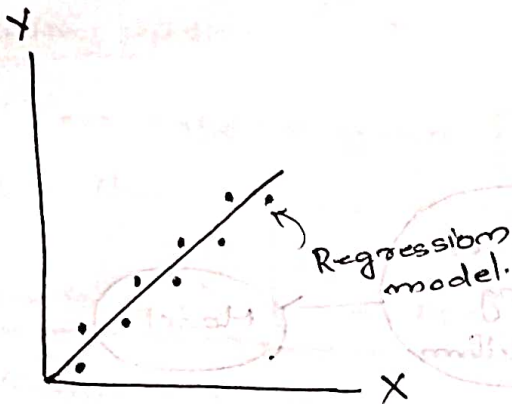
For suitable values for w and w_0

Testing phase

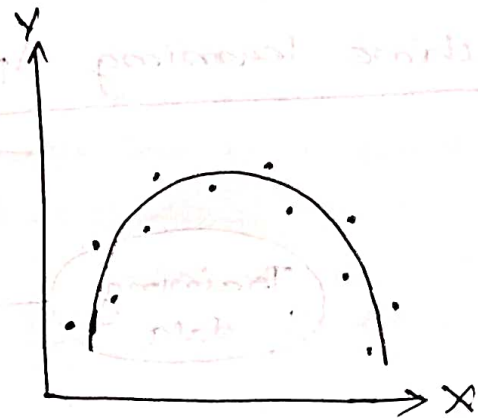


Regression Models

Linear model



Non-linear model



1. Classification
2. Regression
3. Learning Associations

Another application is finding the relationship b/w the data and expressing it as association rule [Association Rule mining]

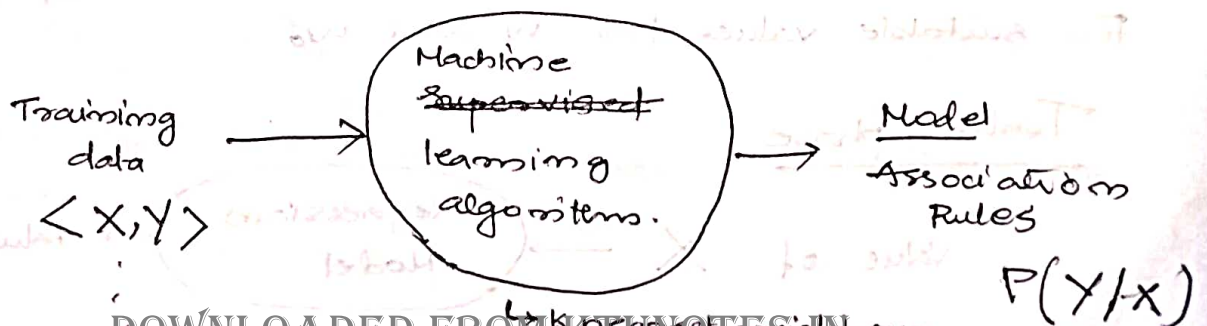
Example:-

Market basket Analysis (Amazon)

The process of analysing customers buy behaviour by finding the association between different items. The customer place is their shopping basket.

Discovering of such association can help retailers (eg:- Amazon, Flipkart) to develop marketing strategies to the customers.

Supervised learning approach



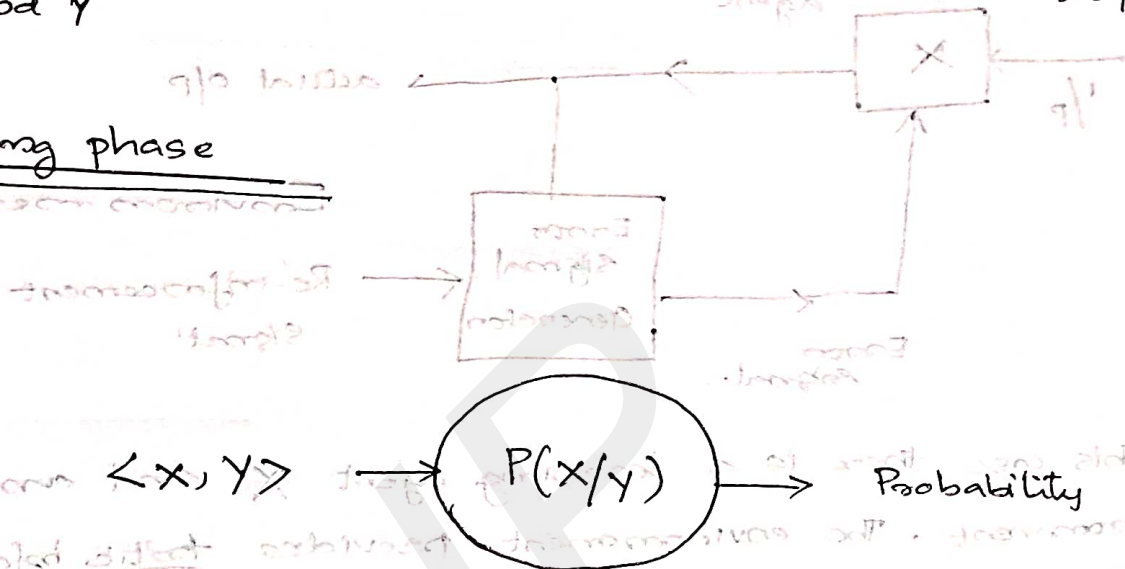
Product or
 X be the set of products a customer already purchased

Y be the product customer buys with X

MACHINE LEARNING APPROACH

Automatically the machine learning algorithm find out the association rule that explore the relationship between X and Y

Testing phase



UNSUPERVISED LEARNING

* There is no supervisor or training phase

* Used for clustering related application.

In a cluster analysis data is partitioned into groups [clusters] based on some measure of similarity and characteristics of data.

Objects in the same cluster should be very similar to each other [intra class similarity] and Objects in different clusters should show dissimilar characteristics. That is inter cluster class similarity should be maximum.

Applications

1) Customer Segmentation.

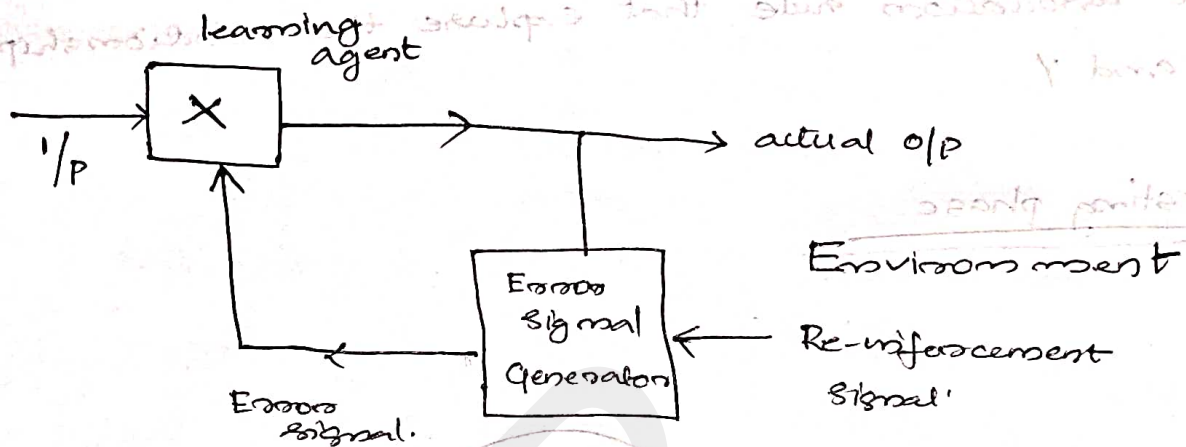
2) document Clustering

3) Image Compression etc

Re-inforcement learning (Robotics)

No supervisor or training

block diagram



In this case there is a learning agent X and an Environment. The environment provides tactic information to produce or correct the actions.

Working

Based on the current state and input the learning agent produce an output. Environment analysis this output and generates the reinforcement signal (information for how to improve the output).

An error signal generate converts this reinforcement signal and feed it back to learning agent X .

based on current I/P and error signal, and state learning agent produces an O/P . the above process repeated untill X produces the decide output. (target)

Applications of RL (Reinforcement learning)

- 1) Game playing
- 2) Autonomous Agents : Robot Navigations
- 3) Adaptive control :

Autonomous cars etc.

Intelligent tutoring etc systems.

Semi supervised learning

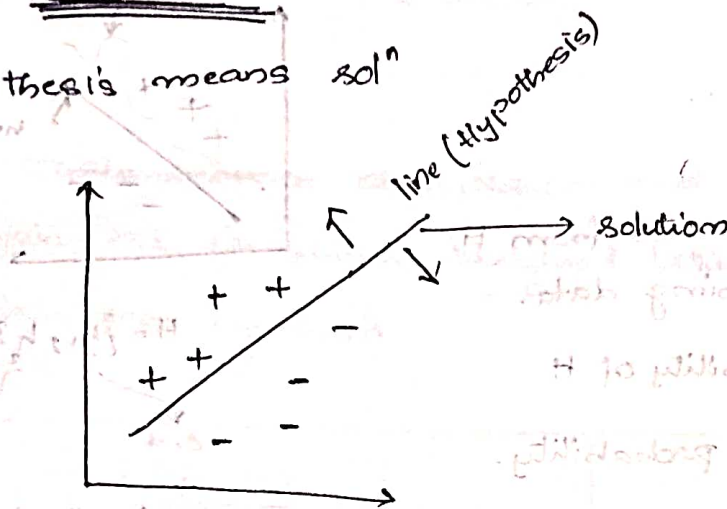
28/09/22

Ktunotes.in

29/09/2022

Hypothesis

Hypothesis means solⁿ



→ H : Hypothesis space

A set of solⁿs to a problem.

→ h : An individual solⁿ.

for example

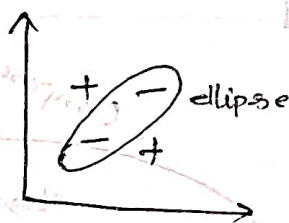
$$h: h_1, h_2, \dots$$

$$h = \begin{pmatrix} h_1 \\ h_2 \\ \vdots \end{pmatrix}$$



$$h = \begin{pmatrix} h_1 \\ h_2 \\ \vdots \end{pmatrix}$$

linearly inseparable problem.



dimension: No of features.

MAP HYPOTHESIS

Maximum a posteriority.

$$P(h/D) = \frac{P(D/h) * P(h)}{P(D)}$$

↓
training data

In machine learning we are interested in finding the probability

$$P(h/D)$$

here,

$h \rightarrow$ A single hypothesis from H

$D \rightarrow$ Observed training data.

$P(h) \rightarrow$ prior probability of h

$P(D/h) \rightarrow$ likelihood probability.

Map hypothesis

Since, there may be more than one hypothesis in H . We can find out the most probable hypothesis $h \in H$ given, the observed data. Such a hypothesis is called Map hypothesis.

$$h_{MAP} = \text{Arg} \left(\max_{h \in H} P(h/D) \right)$$

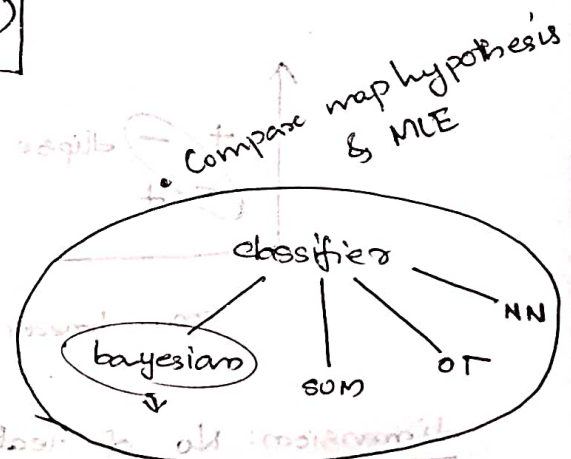
$$= \text{Arg} \left[\max_{h \in H} \left(\frac{P(D/h) * P(h)}{P(D)} \right) \right]$$

$$= \text{Arg} \left[\max_{h \in H} (P(D/h) * P(h)) \right]$$

probability of training data

Baye'sian Classifier

* Uses map hypothesis for classification.



Ktunotes.in

